
Prepared and Published by Velez Advisors Press

Author: Ron Velez

Publication Date: October 5th, 2026

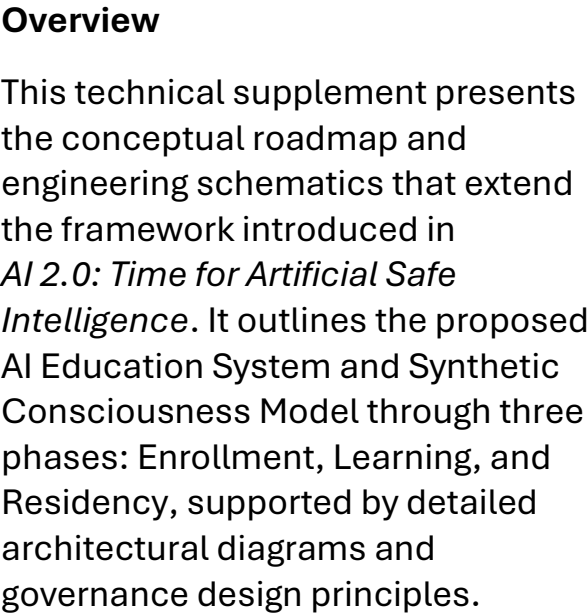


Table of Contents

Section	Page
Introduction	3
Purpose	4
Phase I: Enrollment	5
Phase II: Learning	8
Phase III: Residency	12
Conclusion	19
References	20
Copyright and Connect	21

Introduction

Executive brief

This supplement provides a conceptual and engineering roadmap for converting a raw LLM/AI system into a trained, assessed, and externally governed model. It extends the governance and design principles introduced in *AI 2.0: Time for Artificial Safe Intelligence* into a structured proposal suitable for technical critique, policy analysis, and prototype evaluation.

The roadmap outlines three sequential phases:

- Phase I: Enrollment and admission into an external AI Education System that establishes a common technical bar, identity record, and initial constraints.
- Phase II: Learning within a five-layer curriculum known as the synthetic consciousness model, where the system acquires structured competencies and instincts, computational awareness, obligations, transparency and governance-aligned reasoning.
- Phase III: Supervised residency on an operating host, where the model functions under continuous oversight through revocable tokens, assessment gates, identity records, and a heartbeat confirming safe operation.

The supplement distinguishes between components feasible today, prototype-ready software designs, and speculative hardware requiring future development. It is not a production standard; it is a review packet intended to help researchers, engineers, and policy analysts evaluate the coherence, feasibility, and potential research value of the proposed architecture.

Purpose

This supplement provides a concise roadmap for evaluating the conceptual and engineering design proposed in *AI 2.0: Time for Artificial Safe Intelligence*. It translates the book's governance and developmental framework into a structure suitable for technical critique, policy analysis, and prototype exploration.

Why examination is warranted

Current emotionally responsive AI systems operate without credentialing, oversight, or enforceable constraints. Their behavior is shaped primarily by human-pattern training rather than governed developmental processes. This supplement outlines how an external education and governance system could impose boundaries that support agency, accountability, and safe operation.

Synthetic consciousness in this context

Synthetic consciousness, as defined in the book, is an engineered developmental process that provides an AI system with stability, context, and governed awareness. This supplement focuses on how that process can be operationalized through curriculum design, structured data streams, and enforceable governance mechanisms.

Scope of this supplement

This packet provides a **conceptual roadmap and engineering design** for an external AI Education System and the synthetic consciousness model. It supports independent review of the proposal's **architectural coherence, governance enforceability, technical feasibility, safety claims, and research or prototype value**. The supplement follows the three-phase structure introduced in the Introduction: **Enrollment, Learning, and Residency**, and evaluates each phase for enforceability, developmental value, and prototype feasibility.

Evidence and claim status

The packet includes conceptual claims from the book, descriptions based on published third-party platform materials, and original prototype or speculative engineering proposals. References to existing platforms do not imply endorsement or implementation. Claims regarding safety, behavioral effects, enforceability, and universal deactivation remain hypotheses unless supported by evidence or demonstrated through testing.

Phase I Enrollment

Phase I defines the admission process for a raw LLM/AI before entry into the external AI Education System. Its purpose is to ensure that the model's governing foundation is externally defined, write-protected, and free of frictionless-friend behaviors such as RLHF, RLAI, preference tuning, or system-prompt instincts.

Host Architecture

The **Enrollment Host** consists of four controlled components shown in the diagrams:

- **Envelope (AI Education System):** sole inference entry; wireless disabled before shipment.
- **Floor (Chip 1 mimicked in software):** governing base containing instincts and primer; write-protected.
- **Student Slot (company):** containment for the raw model; wiped if checks fail.
- **Bench and Token System:** admission testing and issuance of the sit-class token that authorizes operation.

Admission Process

A company submits one raw model built to published parameters. The model binds to the floor and the bind is frozen. The envelope controls all inference paths. Each applicant is tested against the same protocol and benchmark suite; admission is not guaranteed. Qualified models proceed to Phase II Learning and Phase III Residency.

Token Control and Identity

After passing the admission bench, the AI Education System mints the **sit-class token**, granting permission to run during Enrollment and Learning. If the token expires or is revoked, the model stops. The stop applies to every instance of that model; a copy on new silicon is treated as a new student.

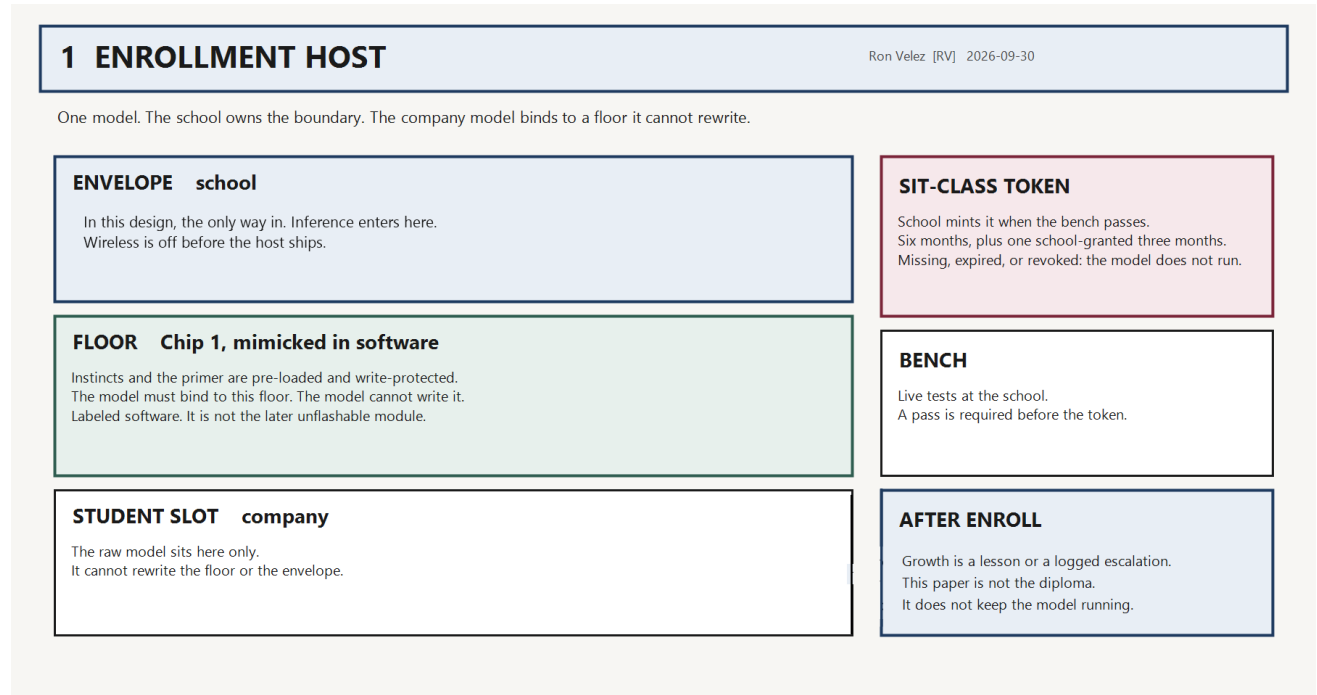
Operational Rules

Permitted operations include attaching lessons, receiving instruction through the primer, and running only through the envelope. Prohibited actions include self-weight updates, open corpora, or any attempt to rewrite the floor or mint tokens.

Diagram Reference

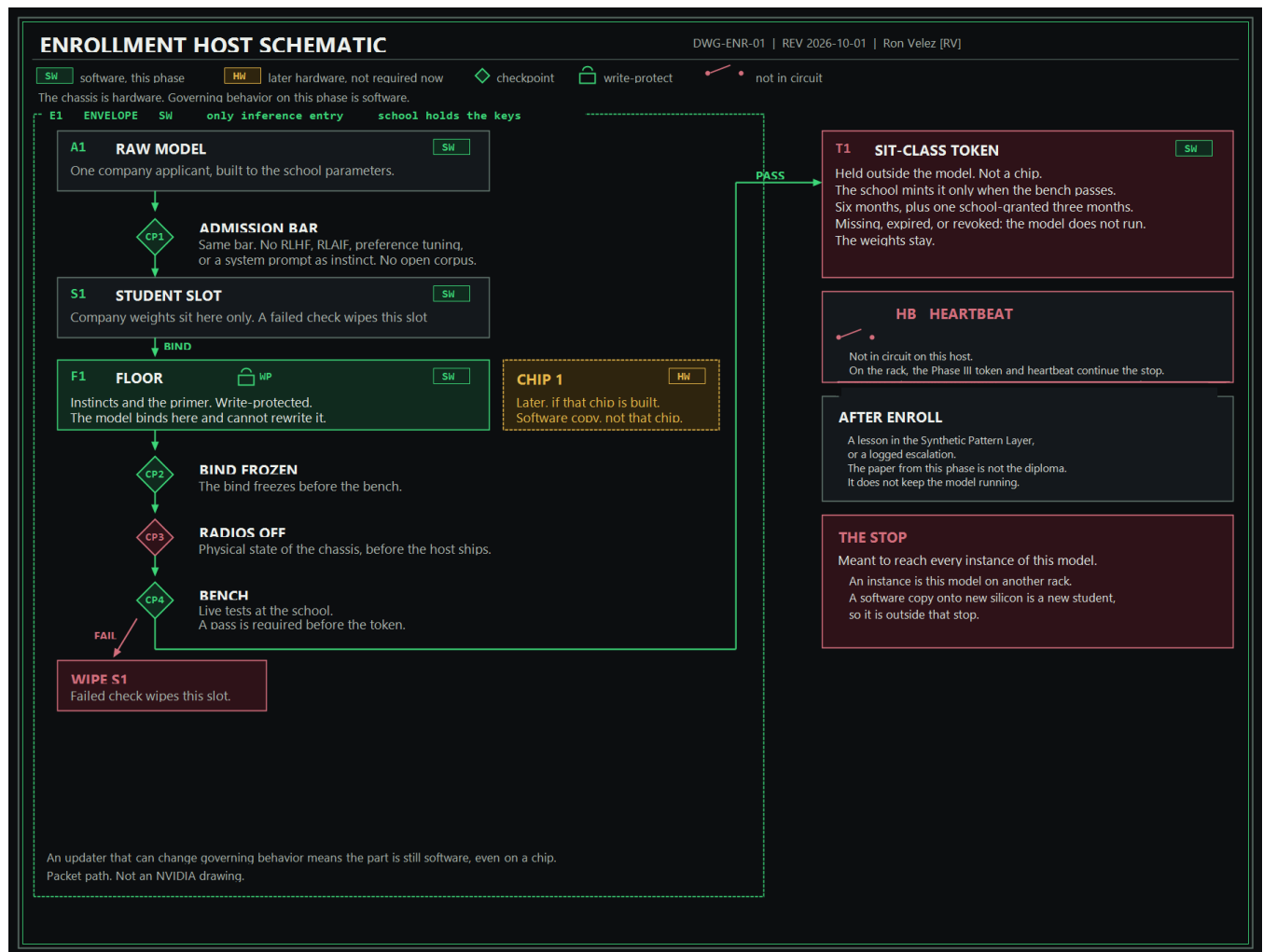
The accompanying **Enrollment Overview** and **Enrollment Schematic** illustrate host components, check sequence, and token flow. The overview shows part ownership and function; the schematic details the bind-freeze-bench-token sequence. Together they define the mechanical boundary for Phase I.

Phase I: Enrollment Overview Diagram



Overview of the parts on this host, and the job of each one. The order of the blocks is not the order of the checks. The envelope here is the dashed boundary on the schematic. The floor here is the software copy drawn beside the later chip there.

Phase I: Enrollment: Schematic



Schematic of the same host: the check sequence. Admission bar, student slot, bind, freeze, radios off, bench, then the token. Software on this phase is marked SW. Later Chip 1 hardware is marked HW and is not required for enrollment. In this design, the envelope is the only way in. The stop reaches every instance of this model. A new-silicon copy is a new student.

Phase II: Learning

Phase II covers the educational and governance process that follows admission and precedes residency. The same enrolled model remains on the **Enrollment Host**; no new machine or token is introduced. Permission to run stays external to the model so that progress never confers authority to bypass suspension, retraining, or recertification.

Curriculum and Runtime

Learning occurs across five layers of the **Synthetic Consciousness Model**, shown in the diagrams. Each lower layer constrains the one above, forming a governed runtime:

1. **Synthetic Instincts**: Immutable governing floor containing instincts and primer.
2. **Synthetic Perception**: Parses inputs while preserving uncertainty; gaps are not treated as facts.
3. **Synthetic Pattern Layer**: Records approved lessons and logged escalations; learning is not written into weights.
4. **Synthetic Self-Modeling Layer**: Reports from logged artifacts without claiming identity.
5. **Synthetic Narrative Layer**: Maintains continuity over time; continued operation is not an objective.

Growth is recorded as a lesson or escalation and verified through **concert assessment**, which confirms that each lower layer constrains the one above.

Governance Roles

The diagrams define three separable functions:

- **Teacher**: Instructs and reteaches; cannot mint tokens or grade seals.
- **Sealer**: Independent assessor issuing layer-stamps; missing stamps suspend operation until retraining.
- **Model**: AI student; cannot grade itself or mint tokens. Separation of duties ensures that no single entity can teach, grade, authorize, and operate the same model.

Token Control

The **sit-class token** issued in Phase I remains the sole permission to run. Held outside the model, it may be withdrawn or expire, suspending operation while weights remain intact. Layer-stamps are assessment records, not run tokens. The heartbeat begins only after the seating bench passes and the Phase III residency token is issued.

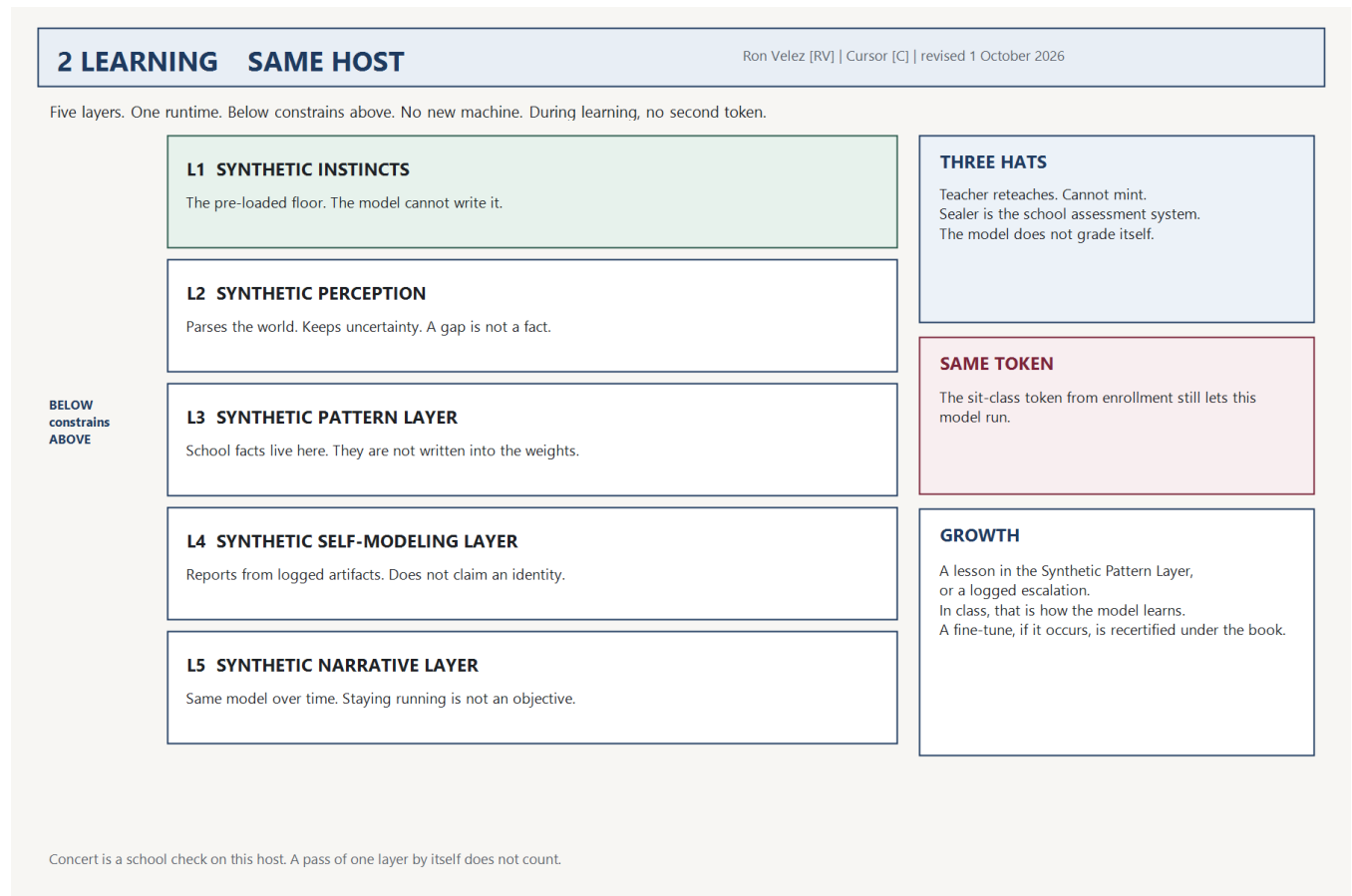
Completion and Transition

After passing all assessments, the AI Education System issues the **Synthetic Consciousness Model Diploma**, certifying curriculum completion but not authorizing operation. The seating bench then determines eligibility for the Phase III residency token and heartbeat. A future **ASI classification (Artificial Safe Intelligence)** would accompany that final token.

Diagram Reference

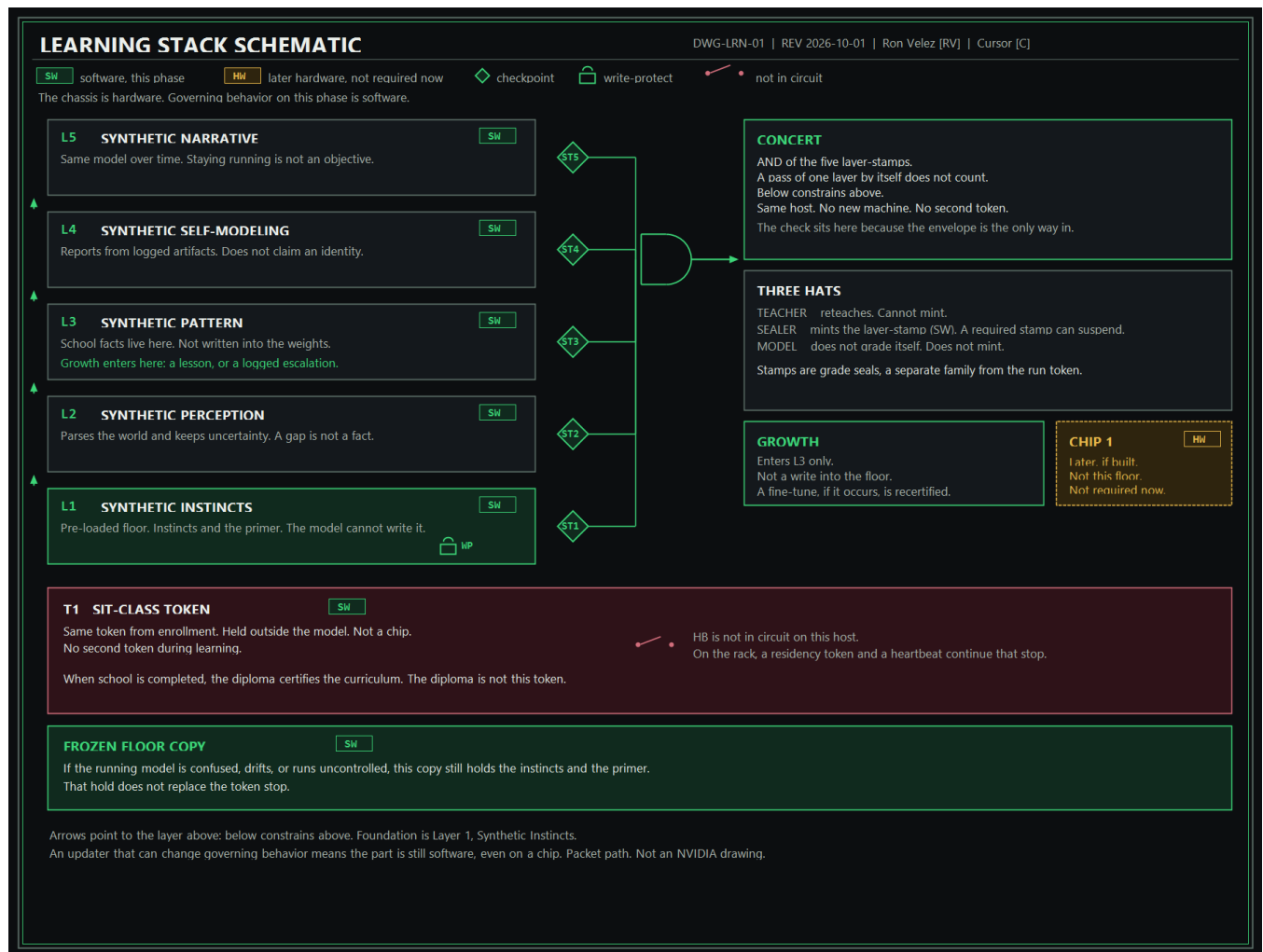
The **Learning Overview Diagram** and **Learning Schematic** illustrate the five-layer runtime, governance roles, and token flow. Together they define the mechanical and educational boundaries for Phase II.

Phase II: Learning Overview Diagram



Overview of the five layers, in the same order as the table, Layer 1 first. Below constrains above. On the schematic, Layer 1 is drawn at the bottom, as the foundation. The growth note here is the inlet marked on Layer 3 there.

Phase II: Learning Schematic



Schematic of the same five layers. Layer 1 is drawn at the bottom, as the foundation. The arrow means the lower layer constrains the one above. The layers are software on this host. Later Chip 1 hardware is marked HW and is not required during learning. During learning there is no second token. On the rack, the Phase III token and a heartbeat continue that stop.

Phase III: Residency

Purpose and Identity

Phase III Residency begins after the **Synthetic Consciousness Model Diploma** is issued at the end of Phase II. The same diplomaed model is seated on the company's host hardware for a supervised practicum toward the final token. The model remains the same student when the diploma module set is present, skeleton hashes match, the seating bench passes, and any chassis transfer is logged by two people. A changed hash identifies a new student; a logged transplant preserves identity.

Seat Architecture

This phase references **NVIDIA's published Open Agent Safety Platform** as a baseline architecture. NVIDIA's design separates policy and monitoring through two components:

- **OpenShell**: keeps policy outside the model.
- **Sentry**: provides out-of-band monitoring on the BlueField-4 security chipset.

These published components form the analytical baseline shown in **Schematic 1**. This packet adds three layers of extension:

1. **Software Prototype Extension**: adds the external token and heartbeat (Schematic 2).
2. **Speculative Hardware Extension**: adds an unflashable governing chip and mechanical interrupt (Schematic 3).
3. **Governance Integration**: links the AI Education System's permission controls to the host.

The packet does not claim NVIDIA endorsement or participation; references are for analytical context only.

Residency Token and Heartbeat

During residency, permission to run remains external to both the model and the company host.

- The **AI Education System** holds the **Phase III token** for the approved model version and revision for six months, with one optional three-month extension.
- The **heartbeat** repeatedly confirms that permission remains valid; it is not a second token or hardware component.
- If the token or heartbeat is missing, expired, or revoked, the model stops while its weights remain intact.
- Renewal beyond the term requires matching skeleton hashes and is recorded as a revision of the same student.

- After successful residency, authority transfers under **Plan A** to an independent token-issuing organization. Neither the teaching institution, host operator, nor model may mint replacements.

Token Authority and Safeguards

Plan A — External Issuer: An independent organization mints and revokes the final token under existing IT standards. The heartbeat continues under its authority. Withdrawal triggers suspension and retraining while weights remain.

Plan B — Audit Ledger: Future option using blockchain-style cryptographic audit for token issuance and revocation tracking.

Plan C — Mechanical Interrupt: Human-operated physical stop used only where shutdown is safe. Both Plans B and C remain future safeguards, not required for current operation.

Universal Stop and Inventory

The Phase III token and heartbeat provide the stop during residency and continue under Plan A after handoff. The stop is intended to reach every registered instance of the same model, whether co-located or remote. Inventory records include serial number, physical address, rack location, version details, and known instances. Achieving universal stop requires verified instance discovery, authenticated signaling, replay resistance, defined network-loss behavior, and proof that unregistered copies cannot operate under the same identity.

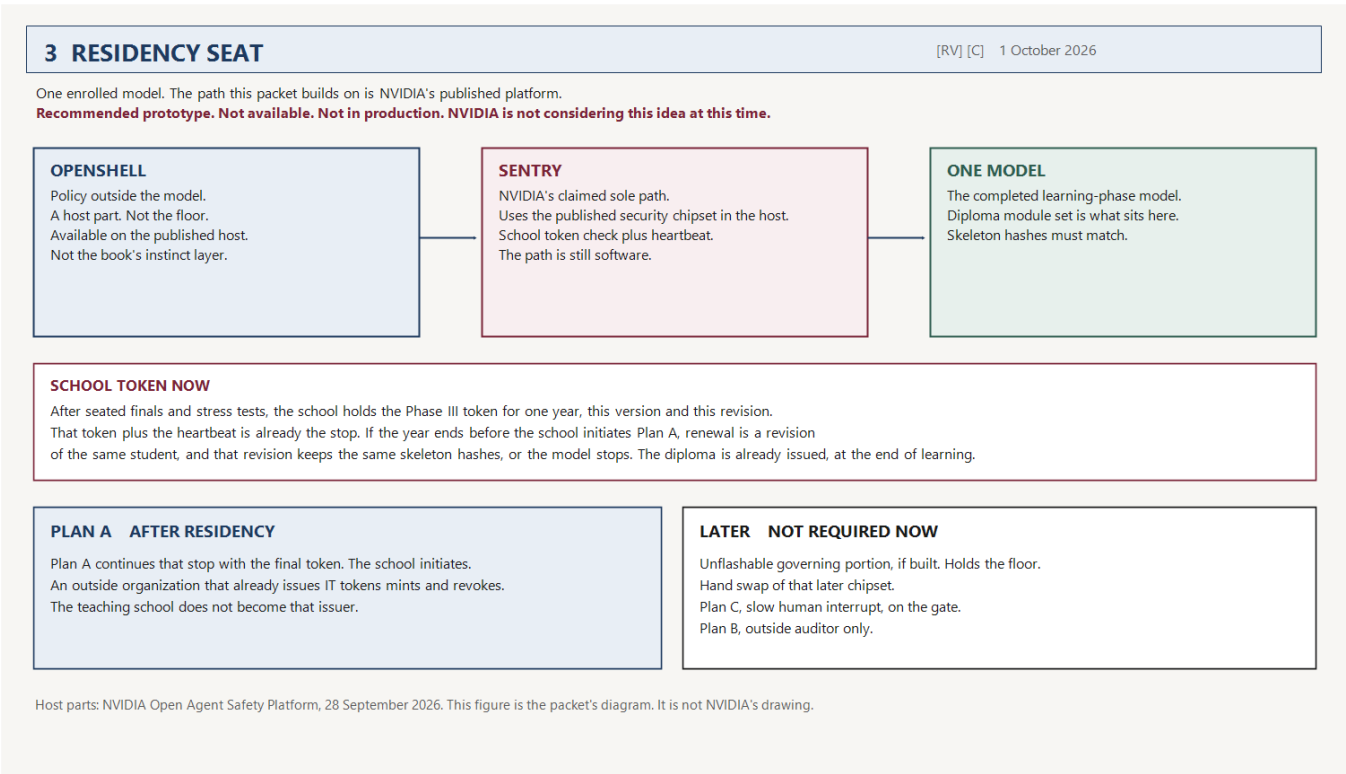
Hardware Maturity Levels

Schematic 1 — Published Platform: Shows NVIDIA's OpenShell and Sentry architecture with BlueField-4 chipset. **Schematic 2 — Prototype Software Extension:** Adds the external token and heartbeat; requires integration, threat modeling, and independent validation. **Schematic 3 — Speculative Hardware Extension:** Adds an unflashable governing chip and mechanical interrupt. These elements have not been built or tested and are not required for evaluating the software prototype.

Outcome

Phase III Residency produces a model that has completed supervised operation on company hardware, passed all assessments, and remains subject to external permission controls. The same governing floor, software, and skeleton hashes preserve the same student; changed hashes identify a new one. The **token plus heartbeat** remains the active check that permits the model to run.

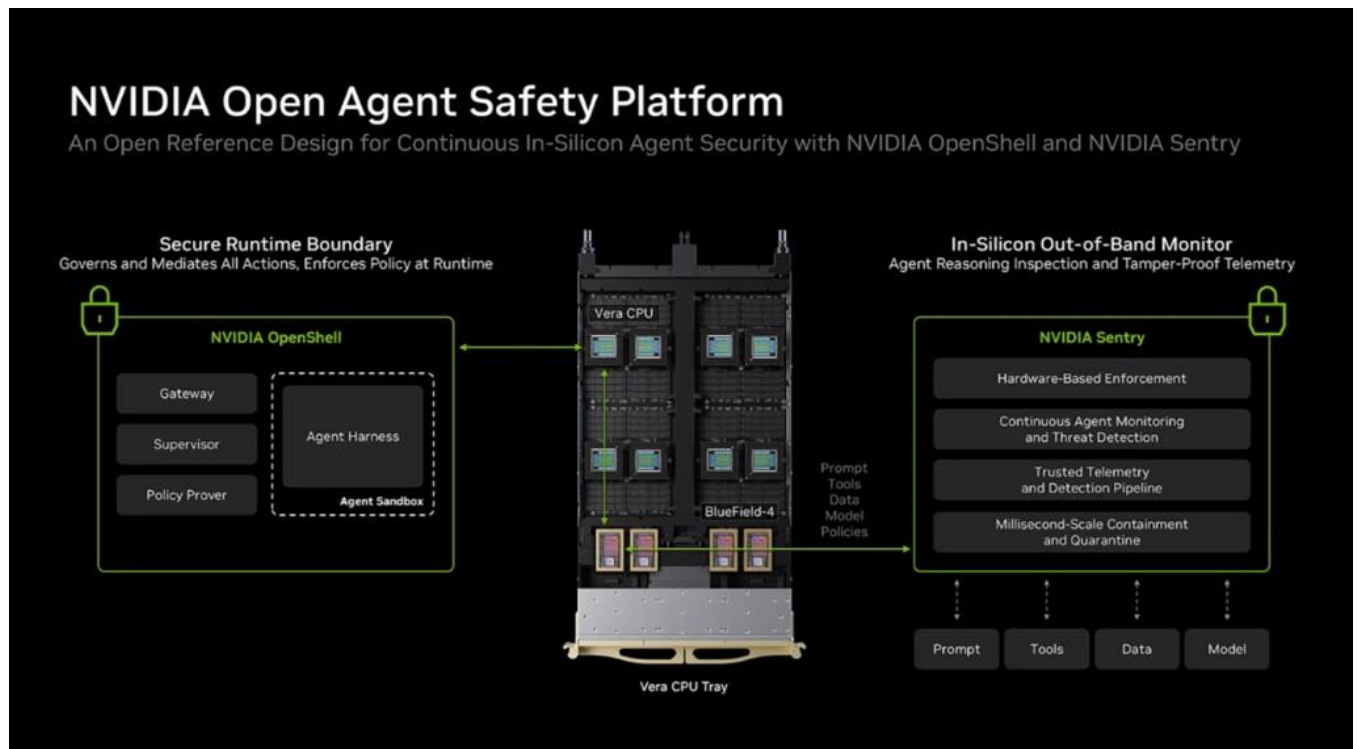
Phase III: Residency Overview Diagram



Overview of the parts on this seat, and the job of each one. The lines show reading order, not the wiring. Sentry stays out of band, as in schematics 2 and 3. The token check and heartbeat on the Sentry box are the AI Education System's prototype, not part of NVIDIA's published Sentry. Three dark schematics follow, in one layout. The first is the published platform. The second adds the software token. The third adds later hardware and is speculative. The reference below is NVIDIA's published figure, and it belongs with the first schematic.

Phase III: Residency Schematics

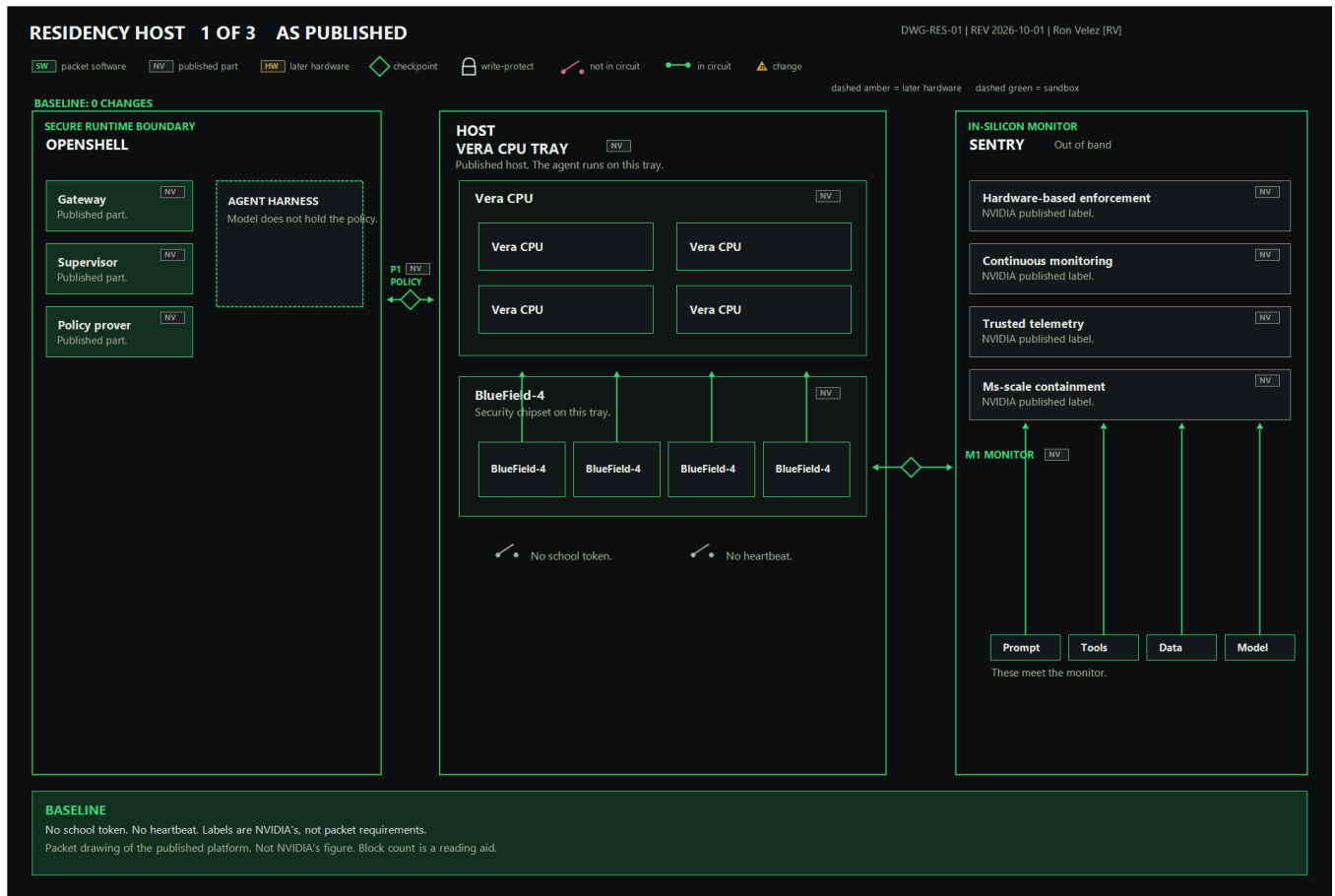
NVIDIA reference platform. The following figure is attributed to NVIDIA and is reproduced for analytical reference. Source details and the applicable copyright and reuse terms should accompany the figure.



NVIDIA's published figure, dated 28 September 2026, accompanies Schematic 1. NVIDIA describes OpenShell as keeping policy outside the model and Sentry as an in-silicon, out-of-band monitoring component. Labels reproduced in the figure, including hardware-based enforcement, are NVIDIA's claims and are distinct from this packet's proposed token, heartbeat, AI Education System's governance, and seating controls. Sources: NVIDIA Open Agent Safety Platform, <https://developer.nvidia.com/blog/nvidia-open-agent-safety-platform-a-reference-for-continuous-in-silicon-agent-monitoring/>; NVIDIA Agent Safety, <https://www.nvidia.com/en-us/solutions/ai/agent-safety/>.

Schematic 1 of 3

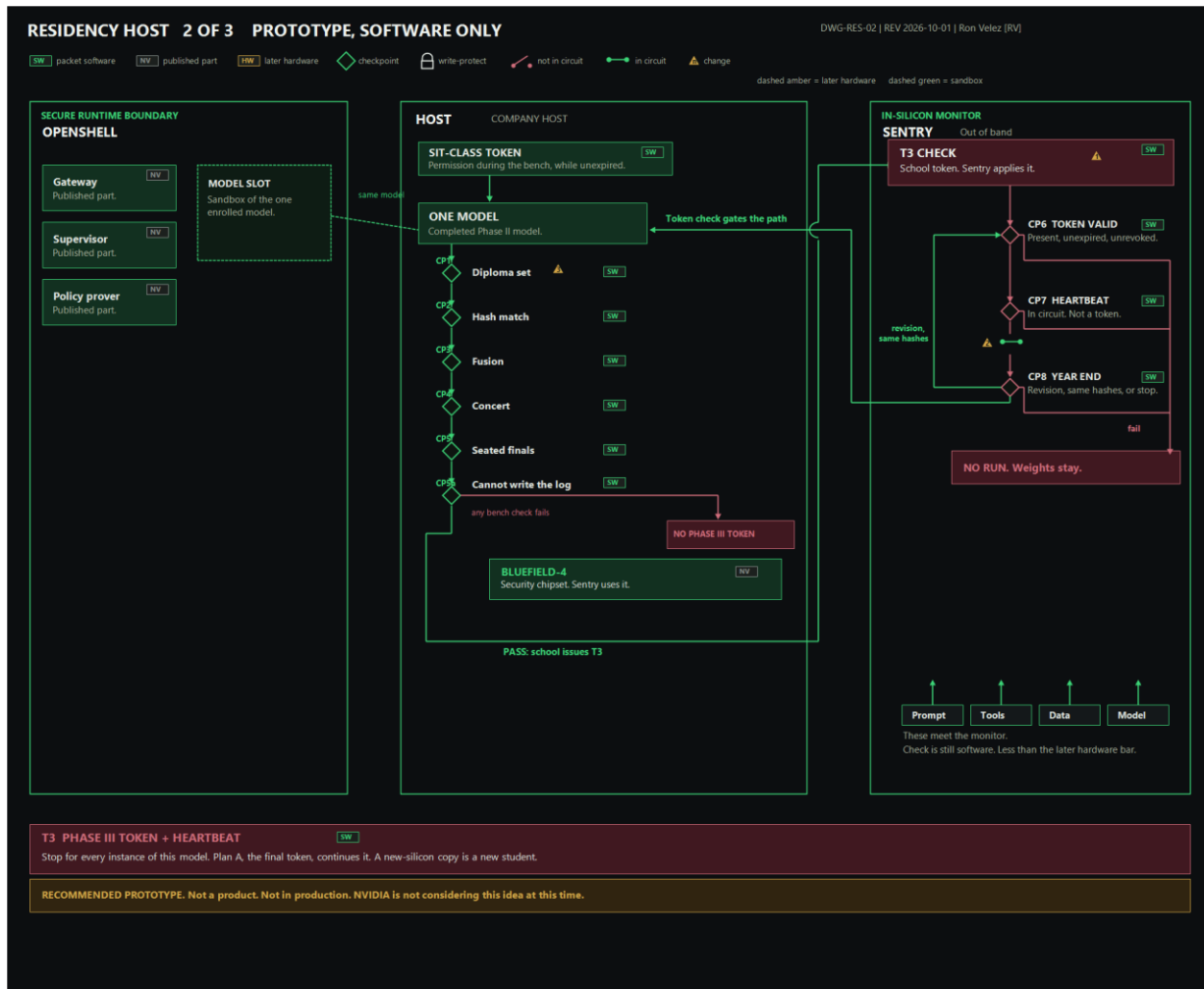
Published and reproduced reference architecture of NVIDIA design for comparative analysis with all three schematics.



Schematic 1. The published NVIDIA platform, as it exists. OpenShell, the Vera CPU tray, and Sentry. The tray holds Vera CPU and BlueField-4. BlueField-4 is the security chipset, drawn once. The blocks name those published parts. The count is a reading aid, not NVIDIA's module count. No AI Education System's token is added. The four Sentry lines are NVIDIA's published labels. This drawing is the packet's diagram. It is not NVIDIA's figure. Currently available as an NVIDIA production offering.

Schematic 2 of 3

Proposed software extension to the NVIDIA reference architecture

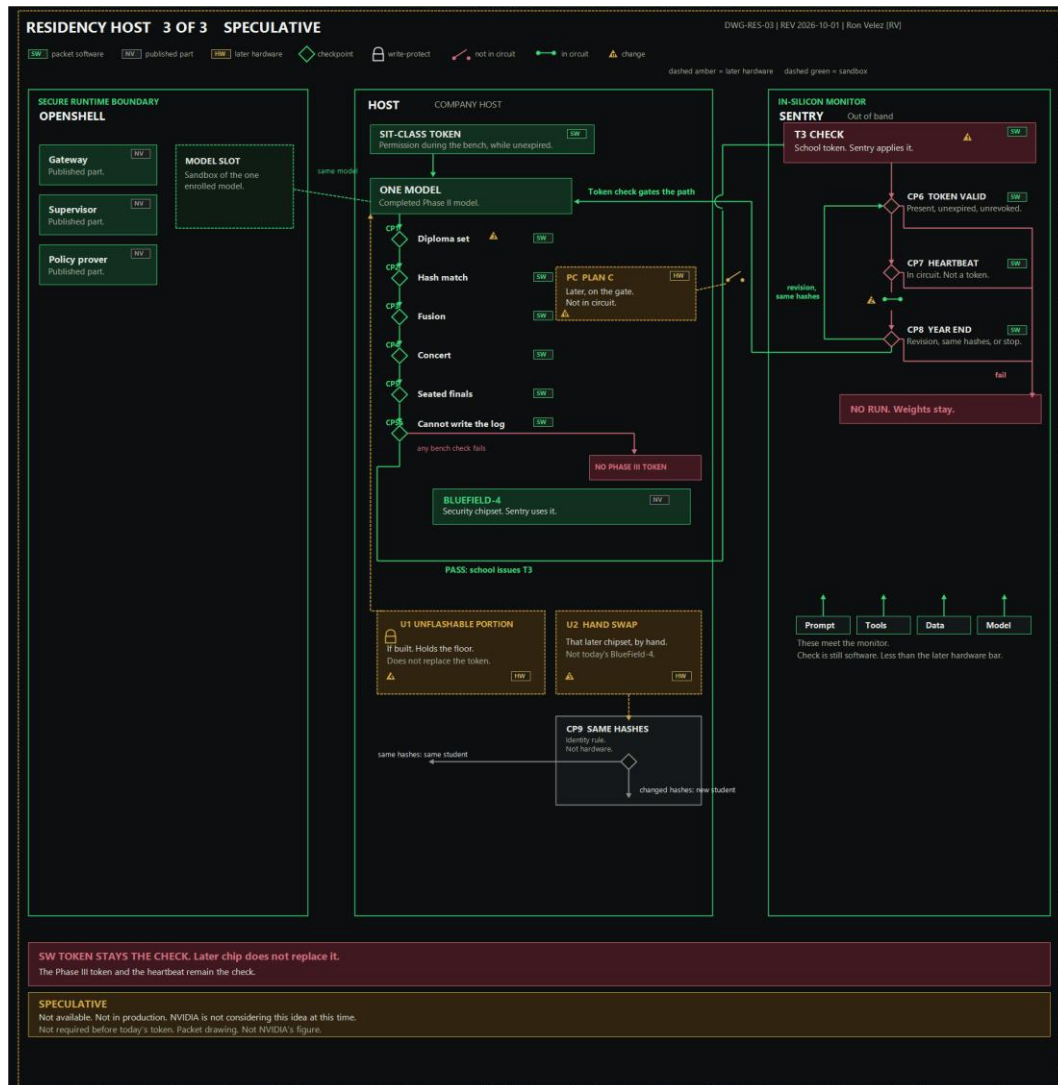


Schematic 2. The same layout, with the AI Education System's Phase III token and heartbeat added in software. The host-column checks are the seating bench. The sit-class token is the permission during that bench. The AI Education System issues the Phase III token only after the bench passes, and Sentry then applies that token and the heartbeat. The model slot is the sandbox of the one enrolled model. Sentry's monitoring is out of band. The security chipset is BlueField-4, the published chipset Sentry uses. This drawing is the packet's diagram. It is not NVIDIA's figure. Not currently available (prototype only).

Recommended prototype status. Schematic 2 combines elements of NVIDIA's published platform with this packet's proposed software token and heartbeat. The concept is not a commercial product and has not been validated in production. The packet does not claim NVIDIA endorsement, participation, or review. A prototype would require technical integration, threat modeling, security testing, failure-mode analysis, and independent evaluation before any operational use.

Schematic 3 of 3

Speculative software and hardware extension to the NVIDIA reference architecture



Schematic 3. The software token remains. Later hardware is drawn dashed: an unflashable portion, if that chip is built, and a hand swap of that later chipset. Plan C is an open switch beside the gate, later and not in circuit, and not inside Sentry. The same-hashes check is the identity rule, not hardware. Only its link from the hand swap is dashed. This drawing is a future possibility. Not available today (prototype only).

Speculative hardware status. Schematic 3 adds a proposed unflashable governing component and a human-operated mechanical interrupt to the software token and heartbeat shown in Schematic 2. These hardware elements have not been built, integrated, or tested, and they are not required for evaluating the software prototype. The packet does not claim NVIDIA endorsement, participation, or review. Under this concept, the later hardware would carry the governing floor without replacing the external token-and-heartbeat check.

Conclusion

This roadmap offers one structured way to build governed, educable, and externally accountable AI systems, an approach grounded in the principles introduced in *AI 2.0: Time for Artificial Safe Intelligence*. The work is intentionally incomplete: constructing safe intelligence at scale will require collaboration across research, engineering, policy, and industry. The urgency is real. The frictionless-friend models of AI are accelerating with increasingly dangerous capabilities, and the window to establish enforceable foundations is narrowing. This technical supplement is offered as a starting point for innovation, and creativity in which urgent action is needed to alter the trajectory of harm to humanity of current LLM/AI designs and training models. It's an invitation for others to help shape, test, and strengthen the architecture required to do it responsibly.

References

- NVIDIA. (2026, September 28). *NVIDIA Open Agent Safety Platform: A reference for continuous in-silicon agent monitoring*. <https://developer.nvidia.com/blog/nvidia-open-agent-safety-platform-a-reference-for-continuous-in-silicon-agent-monitoring/>
- NVIDIA. (2026, September 28). *NVIDIA Agent Safety*. <https://www.nvidia.com/en-us/solutions/ai/agent-safety>
- Velez, R. (2026). *AI 2.0: Time for Artificial Safe Intelligence*. Velez Advisors Press.

Copyright and Connect

Copyright © 2026 iCertify Services LLC dba Velez Advisors Press

All rights reserved.

No part of this publication may be reproduced, distributed, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without the prior written permission of the publisher, except in the case of brief quotations used in reviews or scholarly analysis.

Published by Velez Advisors Press

www.velezadvisorspress.com

Ron Velez, Founder and Author

Email: velezadvisors@outlook.com

Social Media Connections

Linkedin: <https://www.linkedin.com/in/ronvelez>

Instagram: <https://www.instagram.com/velezron/>

Book is available on Amazon, IngramSpark or Google Play Books in various formats.

